

Security Comparison of Blockchain and Cloud-based Identity Management: Considering the Scalability Problem

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Abstract—Identity management (IdM) is used in every facet of our electronic society. The need for IdM and the problems these systems face grow as our electronic identities grow and become more complex. When IdMS is concerned with identifying and controlling the individual and access controls in the system, IdMS faces a new level of complexity as the number of users increases, making these systems vulnerable to various threats and attacks. However, while the concepts of attacker and target are well established in the field of information security, when it comes to privacy, these two concepts can be very limited. Personal information can be used for various purposes, and it is not always the scenario that the privacy threat comes from what is considered an attacker from the beginning. Here comes the Cloud and Blockchain's role in providing more secure and scalable IdMS. This study aims to cover existing solutions by describing their level of security through the risk assessment process to provide a concise picture of the differences.

Keywords—Risk assessment, IdMS, Cloud, Blockchain, Scalability, Privacy

I. INTRODUCTION

Privacy is a concept that has diverse connotations for different people and countries, with what is deemed private in one place not being considered private in another [1]. The notion has been acknowledged since the time of Aristotle and has been a topic of political and intellectual philosophical debate and discussion ever since [2]. However, privacy may be defined as property about an individual's private and sensitive information. In greater depth, privacy may be defined as an individual's, group's, or institution's right to choose when, how, and to what degree information about them is shared with others [3]. Nonetheless, during the last two decades, information technology has swiftly developed beyond these debates, and our concept and protection of privacy are rapidly becoming outmoded. As a result, privacy risks vary depending on the problem and scenario, making defining and recognizing

privacy-related risks difficult. Meanwhile, as people's reliance on online services and technology develops, so does the number of credentials that must be remembered, creating a new complicated sector with privacy problems and increasing the relevance of the identity management system (IdMS) [4]. Identity Management (IdM) and Identity and Access Management (IAM) are interchangeable terminologies and fall within the threat and security assessment services category. Where IdM can be defined as "the management of identity-related information... is simply the digital authentication and certification of identity-related information, with its main application in access management" [5] and "a set of data management systems and practices to increase confidence in the identity of individuals where appropriate" [6]. In other words, it is the

administration of individual identities, including their authentication, authorization, roles, and privileges inside and between organizations. Identity Management Systems (IdMS) have been developed and implemented to assist individuals and businesses in managing their electronic identities, along with many other things [7]. The National Institute of Standards and Technology defines IdMS as "one or more systems or applications that handle the identity verification, validation, and issuing process" [8]. It may be found in all facets of our electronic culture, and many companies now provide IdMS to their clients. However, from a technology standpoint, IdMS is still a novel idea, and the possible privacy intrusions posed by these systems are still unknown. The author in [9] [10] indicated that in today's dynamic world, the IdMS faces several obstacles, particularly with the emergence of big data concepts such as scalability, access control management, customer synchronization, etc. One of the most pressing issues today is scalability, which refers to a system's ability to expand its capacity and usefulness in response to changing user demands. This is especially true in developing rising areas like IoT, Cloud Computing, Blockchain, Social Networks, and Video-on-Demand Streaming industries. On the other hand, new and more substantial vulnerabilities will be imposed, possibly influencing and escalating privacy issues. As a result, consistent and trustworthy identity management solutions that enable effective scaling and usage in any context are required [11]. As a result, several solutions to this problem have evolved, such as commercial software from well-known firms or cloud-based alternatives [12]. However, because blockchain is one of the sectors rapidly facing scaling issues, various research and studies suggest offering Blockchain solutions in other firms and industries [13]. However, as previously said, defining a broad privacy definition is difficult, and precisely assessing privacy issues is similarly difficult. As a result, risk management standards paired with various risk identification and analysis methods will be used to analyze the hazards of occurrences to persons and enterprises to understand these risks better. According to the NIST definition, risk assessment is "The process of identifying risks to organizational operations (including mission, functions, image, reputation), organizational assets, individuals, other organizations, and the Nation, resulting from the operation of an information system. Part of risk management incorporates threat and vulnerability analyses and considers mitigations provided by security controls planned or in place"[14]. This study aims to use risk assessment to investigate and identify the threats, vulnerabilities, Standards and Regulations, and the taken controls [15] of the Blockchain and Cloud computing Identity management system concerning the privacy issues created by inefficient scaling plans in identity management systems. An- answering the main question of this study;

Determining the extent to which cloud computing and blockchain can securely bypass privacy threats while addressing the scalability issue of traditional IdMS.

II. LITERATURE REVIEW

This part outlines relevant research on solutions that compromise privacy and security risks from inefficient scaling plans in identity management systems. Authors in [9] initially reviewed the security concerns associated with identity and access management security in the cloud environment and the strategies presented to address them. In conclusion, several major cloud-IdMS security vulnerabilities were highlighted, including brute-force attacks, man-in-the-middle attacks, Trojan attacks, phishing, and cross-site request forgery. Relatedly, contributors of [16] investigated the privacy issues that arise while using the cloud. In addition, it investigates the legal safeguards to secure organizations participating in cloud computing. According to the findings, most firms that used cloud services solutions needed more technical safeguards for identity management and authentication, a lack of privacy awareness and legal sanction, and an insufficient departure strategy, data assurance, and portability. While in [17], a novel, more secured IdMS has been proposed to perceive the privacy and data transmission security in the cloud environment in a framework that compromises attacks, such as the replay attack. In the same point of view, researchers of [18] introduced the security and privacy challenges that necessitate an adaptive solution strategy consistent with current and future cloud security. The study analyzed many studies in the literature, considering its adaptability in mitigating future recurrent risks and demonstrating how cloud security conflicts have invalidated their offered models. In [19] continuing what was previously introduced and using the STRIDE technique, the essay highlights the security vulnerabilities associated with cloud computing from the user's standpoint. Furthermore, it analyzes many ineffective methods in the literature and recommends establishing a safe, adaptable cloud environment. [20] used the cloud and identity management available literature to examine the trend toward future research in this domain. From this context, [21] presented security measures to ensure privacy in cloud services, particularly in IdMS. The author also recommended methods for retaining user anonymity while maintaining user privacy, such as designing an architecture that ensures security and privacy while utilizing customized cloud systems. While [22] 's technique allowed users to conceal their identities throughout data collecting. [23] developed a strategy for ensuring user privacy while detecting anomalous intrusion and detection in-network traffic using Homomorphic Encryption (HE). [24] reviewed the various cloud computing risk assessment models in use, comparing and rating their strengths and weaknesses, and found that quantitative and qualitative risk assessment models must be better suited for the cloud computing environment. Consequently, they proposed a new risk assessment model that fully handles all elements of cloud computing that the previous models did not address. [25] developed a Risk Adaptive Access Control (RAAdAC) in the same context for safeguarding the privacy of sensitive data in flexible real-time. [26] suggested a RAAdAC system that was a mix of Policy-Based Access Control, Attribute-Based Access Control, and Machine Learning. To do this, the authors established guidelines for measuring risk, establishing an acceptable level of risk, and ensuring that all information was customized to stay within the acceptable

level. Compared to traditional access control, RAAdAC offers adaptation and flexibility by adhering to these principles. By verifying previous access control choices and then quantifying the privileged, subject, and object, the RAAdAC system detected problems when access was sought. [27] provided a paradigm for modeling how users interact to attain goals. The suggested architecture used a catalog to help software programmers through the many options for ensuring privacy. The author of [28] illustrates how to strategically reason about non-functional privacy needs by permitting only modeled user interactions. The framework's strength is its adaptability, which allows it to attain goals from various choices.

The contributors in [29] concentrated on the Blockchain solution, describing how it mitigates the risks associated with several identity management systems difficulties such as scalability, interoperability, mobility, security, and privacy. [30] presented a blockchain-based, safe, and decentralized architecture to address security and privacy concerns such as denial of service and man-in-the-middle attacks. Furthermore, the suggested system was verified using a case study to inspect the open research issues related to the blockchain-based IdMS. In the same vein, [31] offered a study of the various risk estimate strategies and compared the various risk-based access control models in terms of risk estimation methodology, risk variables, and assessment domain in upcoming Blockchain technologies. [32] also proposed alternate models for effective Blockchain-based identity management and authentication, as well as some concerns for policymakers, regulators, enterprises, and people when dealing with privacy issues and risks. [33] presented the BIdM, a decentralized identity management system based on blockchain, to address the security and scalability issues existing centralized identity management systems are experiencing. According to feasibility and performance evaluations, the BIdM delivers outstanding authentication optimization compared to conventional identity management systems. However, each approach has its own set of advantages and disadvantages. In 2021, Sung and Park used an academic literature analysis and an actual design case designed for the government sector to investigate the benefits and obstacles of adopting a blockchain-based identity management system. Concluding that, Blockchain-based identity management systems may increase security, openness, accountability, and dependability in user ownership of one's data while lowering the time and cost required to supply services and enhancing administrative efficiency. However, such systems are only sometimes straightforward to deploy, and implementing new technology in any sector is a difficult and time-consuming procedure. [34] Discuss the existing security standards utilized in Blockchain systems and future potential difficulties by examining the IIoT architecture. Furthermore, the study examines the criteria established in industry standards for a secure system. In the context of BC solution, [35] Analyzed the different PII given to users to validate existing blockchain-based identity solutions, using data from real cases of theft and attack stories to provide a method for a truly effective solution, helping users make informed decisions, and incentivizing providers to offer better options to their users. [13], provided a taxonomy to categorize the different blockchain architectures and models, highlighting the security consideration. While [36] analyzed the decentralized blockchain IdM through the definition of

topology patterns and studied the trust requirements for these patterns to compare these two models.

III. RESEARCH METHODOLOGY

A four-step risk assessment demonstrated in figure 1 below will be conducted to answer the main question of "Determining the extent to which cloud computing and blockchain can securely bypass privacy threats while addressing the scalability issue of traditional IdMS" from four perspectives of the threats, vulnerabilities, standards, and regulations that protect consumer rights and the risk mitigation controls by answering questions such as "How likely each technology will be attacked?" "Which is the most secure?" and "What are the standards regulating each technology?"

- **Threats Identification:** this section will identify the threats and hazards related to the technology that may face the company when one of the two solutions is implemented (Cloud-based or Blockchain-based IdM)
- **Vulnerabilities Identification:** this section will identify the weak points that an attacker may use to breach the system's security
- **Standards and Regulations:** this section will highlight the International standards that assist and advise providers regarding the security measures required to reduce attacks. As a result, enterprises selected technology that matched their anticipated security needs. If the user isn't given concrete criteria, they won't be able to make an informed judgment about the identification solution. These solutions and the options available may overwhelm them and perhaps negate the objective of delivering a more secure identity solution without scientifically produced practical instructions.
- **Risk Controls:** this section will outline the safeguards employed by each technology to minimize and limit the possibility and effect of potential assaults.



Fig. 1. Risk Assessment process

A. Blockchain

The Threats and Vulnerabilities discussed in this section are drawn from real-world examples in several research articles.

➤ Threats Identification

- **Compromise user private key:** due to the lack of the required level of randomness in the public key in the

Elliptic Curve Cryptography (ECC) technique, if the same random value (nonce) is used in more than one signature, making it possible to compromise the users' private keys. [37]

- **Preimage and collision attacks:** exposing the cryptographic primitives used
- **to assure the validity and precision of the operation** [38]
- **Hash Function Vulnerability:** To assure the validity and precision of the operation, certain blockchain networks rely on cryptographic primitives. With the fast advancement of computer power and improved cryptanalysis, these primitives have been exposed to various cybersecurity concerns. [39]
- **Mining Malware Crypto-jacking** uses a significant amount of electricity and
- **may damage the target's system functioning and compromise some of the access controls on the system.** Attack Control Blockchain Network: the 51 percent Vulnerability means that attackers might gain control of the blockchain network by exploiting the mechanism's 51 percent vulnerability. [40]
- **Time-Jacking Assault:** A special attack on the BC network in which full nodes preserve network time. Consequently, if the attacker is connected to the target node, it will immediately attempt to divulge wrong timestamps, which will modify the node network time clock, isolating the target node from the network or decreasing the transaction rate on the whole network. [37]
- **Denial-of-Service Attack:** even with the blockchain being a peer-to-peer (P2P) technology, it is still vulnerable to such attacks, especially DDos. [41]
- **Sybil Attack:** The adversary sets up phony assistance nodes to disclose a portion of the network and isolate the targeted node via a group of nodes or merely blocks. This may result in false information to the network state's intended recipient. As a result, a successful assault can disable the consensus algorithm's operation and result in a potential double-spending attack. [41]

➤ Vulnerabilities Identification

Table 1 shows the vulnerabilities in the blockchain that might allow an attacker to exploit the system [15, 37, 38, 40, 41].

Table 1. Blockchain Vulnerabilities

Threat	Vulnerability
Exposing the user's private key	Digital Signature Vulnerability, Insufficient ECC
Preimage and collision attacks	Hash Function Vulnerability
Mining Malware Crypto-jacking	Malware of PC
Attacker Control Blockchain Network	51% Vulnerability
Time-Jacking Attack	An infected node in the network
Distributed Denial-of-Service Attack	Insufficient network bandwidth
Sybil Attack	Malicious node

➤ Standards and Regulations

ISO In 2016, the ISO/TC 307 Technical Committee introduced the project of blockchain and distributed ledger technologies to standardize blockchain and distributed ledger. [42] Since then, several standards and projects relating to the BC have been launched, including ISO/TR 23244, [43] which was published in 2020 and provided an overview of privacy and personally identifiable information (PII) protection as applied to blockchain and distributed ledger technologies (DLT) systems. Most crucially:

- Privacy policies
- Privacy and Identity management
- Privacy controls
- Privacy impact analysis as part of the overall risk management, including Threats, vulnerabilities, consequences, and risk mitigation
- Privacy management in blockchain and DLT

Furthermore, the ISO is working on the ISO/PRF TR 23249, which will offer an overview of existing DLT systems for identity management. [44] NIST experts have investigated blockchain technology, including use cases, applications, current services, protocols, security assurances, and cryptographic techniques. The research achievements include scholarly articles and the development of software for experimentation, as well as giving guidance for other NIST efforts in this area. As a result, one of NIST's new Blockchain initiatives is NISTIR 8202, the Blockchain Technology Overview. This document gives an overview of blockchain technology at a high level. The goal is to provide readers with an understanding of how blockchain technology works. [45] Moreover, NISTIR 8301 undertakes Blockchain Networks: Overview of Token Design and Management. At the same time, the OWASP Top Ten is meant to guide developers on the most frequent security flaws in online development. Even though blockchain systems are not standard web apps, many of the same risks apply. As a result, [46] have created a simple mapping between the OWASP The top Ten list of web application vulnerabilities and the blockchain realm is in the points below.

- Injection
- Broken Authentication
- Sensitive Data Exposure
- ML External Entities (XXE)- not directly applicable to some component of the blockchain
- Broken Access Control
- Security Misconfiguration
- Cross-Site Scripting (XSS)
- Insecure Deserialization
- Using Components with Known Vulnerabilities
- Insufficient Logging and Monitoring

GDPR While the GDPR's standards foster openness and trust between data subjects and companies, organizations that keep personal data on permissioned blockchains breach them since blockchain's basic

immutable ledger technology prevents data subjects from exercising their rights. As a result, there is a clash between Blockchain and GDPR. [47] The tension between the GDPR and blockchain-based data privacy methods stems from two fundamentally divergent perspectives on effectively preserving data privacy. Blockchain argues that sophisticated encryption and distributed ledger systems of storage and protection that are decentralized and immutable are the best ways to preserve privacy rights. It is meant to be a permanent, tamper-proof record independent of any governing entity [48] [49].

➤ Risk Control

Some of the controls implemented to mitigate the risk's impact and likelihood could be summarized as follows: [45, 50, 51].

- Blockchain is adding authentication and permission rules to offer authentication, authorization, and encryption for data access. By applying cryptographic techniques, which helps maintain confidentiality.
- Use threat intelligence, a high-level approach for acquiring important information about a new or current cyber threat.
- Sequential hashing and encryption make data manipulation nearly impossible, and immutability and transparency maintain data integrity.
- Another use case where blockchain might be used is to reduce the impact of DDoS assaults, making these attacks less common. DDoS assaults are becoming expensive since they attempt to overwhelm the network with a massive volume of tiny transactions.

B. Cloud Computing

➤ Threats Identification

Privacy-Threats that could face the cloud environment could be summarized as; [12, 12, 15]

- Ransomware: is a type of malware in which the attacker utilizes encryption to encrypt files or other resources on the victim's machine as it encrypts any data attached to a computer.
- Human Error: Passwords may be obtained, and cloud accounts hijacked by hackers via fraud, phishing, and social engineering forms
- (APT) Advanced Persistent Threat: a form of abuse where an attacker penetrates the systems to gain a foothold in the enterprise's cloud infrastructure to access and steal the data. APTs typically alter the security measures employed to counter them as they follow their targets surreptitiously for lengthy periods.
- Shared Technology Vulnerabilities: Several leases are possible in cloud computing, allowing multiple users to share different cloud resources. The underlying infrastructure components that enable cloud service deployment may not be built to provide robust isolation features for multiclient applications or multitenant architecture. By exploiting configuration error or a vulnerability in the cross-platform component, an attacker might jeopardize the cloud data security of many or all clients, resulting in a data breach.

- **Account Hijacking:** the theft of a genuine user's account credentials and their use for malicious reasons. Attackers using stolen credentials may jeopardize the CIA of cloud services.
- **Authentication and Identity Management:** Organizations frequently struggle with managing identities as they attempt to provide the proper rights to each position of the user's job. Cyber hackers accessed millions of personal information records due to a data breach.
- **Vulnerable Systems and APIs:** The PaaS-level OS may contain exploitable vulnerabilities in system software that allow an attacker to disrupt service operations or take control of the system. In addition, an IaaS-level hyper-visor may also contain exploitable vulnerabilities to breach the availability, confidentiality, and integrity of all clients on one or multiple machines.
- **Data breach:** Data breach is a high-risk issue and ranks number one among the risks of cloud computing. A data breach is a security event in which an unauthorized entity accesses, copies, or transmits confidential, sensitive, or private data concerning an organization or a person.
- **Denial of Service:** A DoS (Denial of Service) attack impacts system availability. In a DoS assault, one source device only originates from the dress and is subject to mitigation. DoS attacks are intended to prevent accessing data or apps from legitimate service users.

➤ Vulnerabilities Identification

Table 2 shows the vulnerabilities in the cloud that might allow an attacker to exploit the system [12, 52, 53].

Table 2. Cloud Computing Vulnerabilities

Threats	Vulnerabilities Areas
Ransomware	Platform
	Infrastructure
	application
Human Error	Insufficient SETA programs
	Human Negligence
Advanced Persistent Threat (APT)	3rd Party API
	Hacking
	Malware
	Phishing
DoS	Network-related issues
Weak Authentication and Identity Management	Social Engineering
Account Hijacking	Man-In-The-Middle (MITM)
Data Breaches	MITM
	Malware
	Human Error/ 3rd party error
	Simple or Poor Security Policies
	Targeted Attack
	Application Vulnerabilities
Vulnerable Systems and APIs	Improper handling of IdMS permission
	Misconfiguration
	Hypervisor Bugs Unpatched Software
Shared Technology Vulnerabilities	Weak API Credentials
	3rd Party vulnerable SW
	Hypervisor vulnerabilities

➤ Standards and Regulations

ISO In 2015, ISO produced the ISO/IEC 27017 code of practice for information security controls applicable to cloud services, based on the ISO/IEC 27002:2013 code of practice for information security standards [54]. Furthermore, ISO published the ISO/IEC 27018:2019 standard, which addresses one of the most critical aspects of cloud computing privacy: protecting personally identifiable information (PII) [55]. This standard focuses on security policies for public-cloud service providers who process personally identifiable information in two ways. This resulted in fully new personal data security measures.

NIST In 2011, NIST released the Special Publications 800-144 standards regarding the Security and privacy challenges in public Cloud-computing and provided some recommendations for the companies to overcome these challenges, [56] where NIST identified nine privacy issues in the Cloud environment, as follows:

- Governance
- Compliance
- Trust
- Architecture
- Identity and access management
- Software Isolation
- Data protection
- Availability
- Incident response

Furthermore, in 2020, NIST issued the SP 800-210 draft under the title "General Access Control Guidance for Cloud Systems," considering cloud problems by assessing the access controls of the three cloud delivery services – IaaS, PaaS, and SaaS [57]. OWASP OWSAP published as one of its top 10 series, the top 10 cloud security risks with a recommendation to mitigate such risks, where the top 10 risks listed were [58]:

- Accountability and Data Ownership
- User Identity Federation
- Regulatory Compliance
- Business Continuity and Resiliency
- User Privacy and Secondary Usage of Data
- Service and Data Integration
- Multi-Tenancy and Physical Security
- Incident Analysis and Forensic Support
- Infrastructure Security
- Non-Production Environment Exposure

GDPR To establish required controls and standards for compliance, the cloud service provider must specify its function by the GDPR rules because deciding on a role is crucial. It makes finding the GDPR obligations that apply to the service provided simple. As a result, defining roles and duties is the first stage in building a suitable strategy for protecting the data. Proactive steps toward building a plan for a data protection plan for GDPR adoption are expected

to be taken by the Cloud service providers as follows [59]:

- Develop guidelines for personal data processing processes.
- Create a data processing procedure and enforce the rights of data subjects, such as the right to access, modify and obtain their information, the right to withdraw consent, and the right to object to the processing activities of the cloud service provider and after.
- Establish privacy criteria for people involved in data processing and control operations from the start.
- Create and implement rules for data portability rights and ownership.
- Put in place data-privacy-protection security measures.
- Develop principles for transferring personal data to international partners and foreign countries.
- Create rules and processes for dealing with breaches and occurrences.
- Create policies for forming commercial agreements, data retention periods, and other related needs.

Risk Control

Some of the controls implemented to mitigate the risk's impact and likelihood could be summarized as follows [21, 60, 61]:

- Appropriate rules should be established for data replication and backup, and there should be reviewable evidence of the adequacy of recovery operations, such as complete, accurate, and rapid data recovery.
- Providing users with security education and certifications, establishing acceptable usage standards, and implementing other security best practices.
- To protect the cloud infrastructure from this risk, it is necessary to provide advanced protections pro-, regular tracking of the asset infrastructure, and robust process management.
- Using best client deployment and data management practices helps reduce common technology vulnerabilities.
- Account credentials should not be shared between cloud services and users, and multifactor authentication should be used wherever possible.
- Controls must be implemented at the network level to protect systems and data from unauthorized use, disclosure, corruption, or loss.
- – Limit the sharing of account credentials between cloud services and users, and if feasible, utilize multifactor authentication.
- Service providers must demonstrate the effectiveness and effectiveness of the firewall setup, provide auditable evidence of the adequacy of access rights, and implement only authorized changes. The information stored in the cloud must be recognized and suitably classified. Cloud providers must show appropriate data privacy measures to customers.

- The cloud service provider's security and information workers must have appropriate expertise and abilities to promptly prevent, identify, and respond to breaches.
- Third-party audits must be performed regularly to evaluate the cloud provider's adherence to the agreed-upon conditions, to ensure adherence to procedures, standards, and policies, and to verify that no substantial changes happen in any such processes or policies.

IV. CONCLUSION AND FUTURE WORK

To summarize, the cloud is now the safest choice from a logical standpoint, while Blockchain ambiguity is far more effective from a security and privacy standpoint. However, it is impossible to determine which technology is better in an absolute sense and without considering how, as with everything else in business, the answer will vary based on the organizational needs and specifications. Eventually, we can conclude that the two systems are excellent scalability possibilities. While from a security standpoint, Blockchain appears to be more effective, as it circumvents the cloud's risks and vulnerabilities. However, Regarding the emphasized blockchain technologies and terminology, there appears to be a lack or delay in establishing a clear set of rules and laws. The lack of international standards and the complexity of its infrastructure make it hard to ensure confidentiality, availability, and integrity, which could result in network end-user safety being jeopardized. Sensitive information being exposed to the public.

As a result, surveys and interviews will be conducted in the future to determine the number of enterprises that have used the blockchain solution, as well as the security issues and attacks that they have experienced.

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